

APACC: Asian Pacific American Community Center

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Table of Contents

**01 Introduction
Appointment**

**02 Process &
Redesign**

**03 Daycare Application
Process & Redesign**

**04 Tables & Database
Diagram**

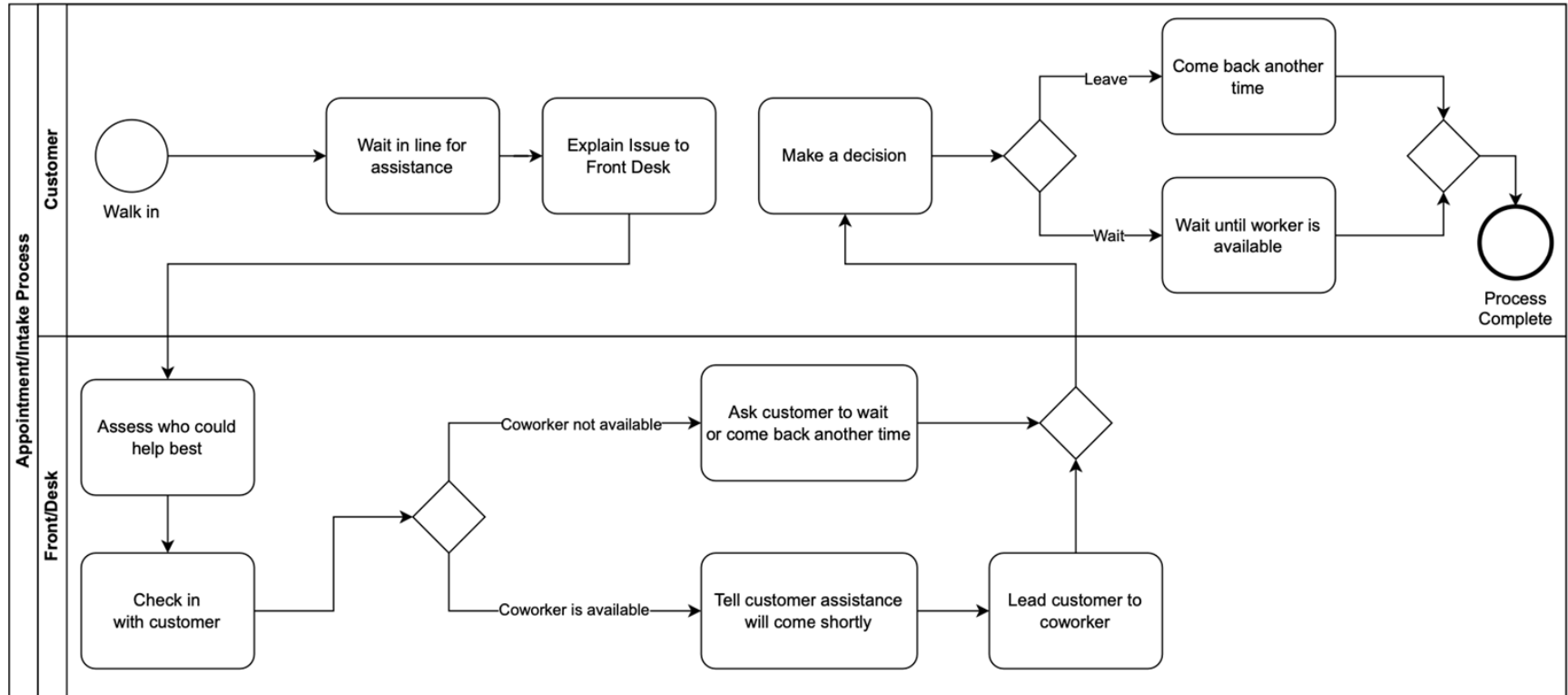
05 SQL Queries

06 Gantt Chart

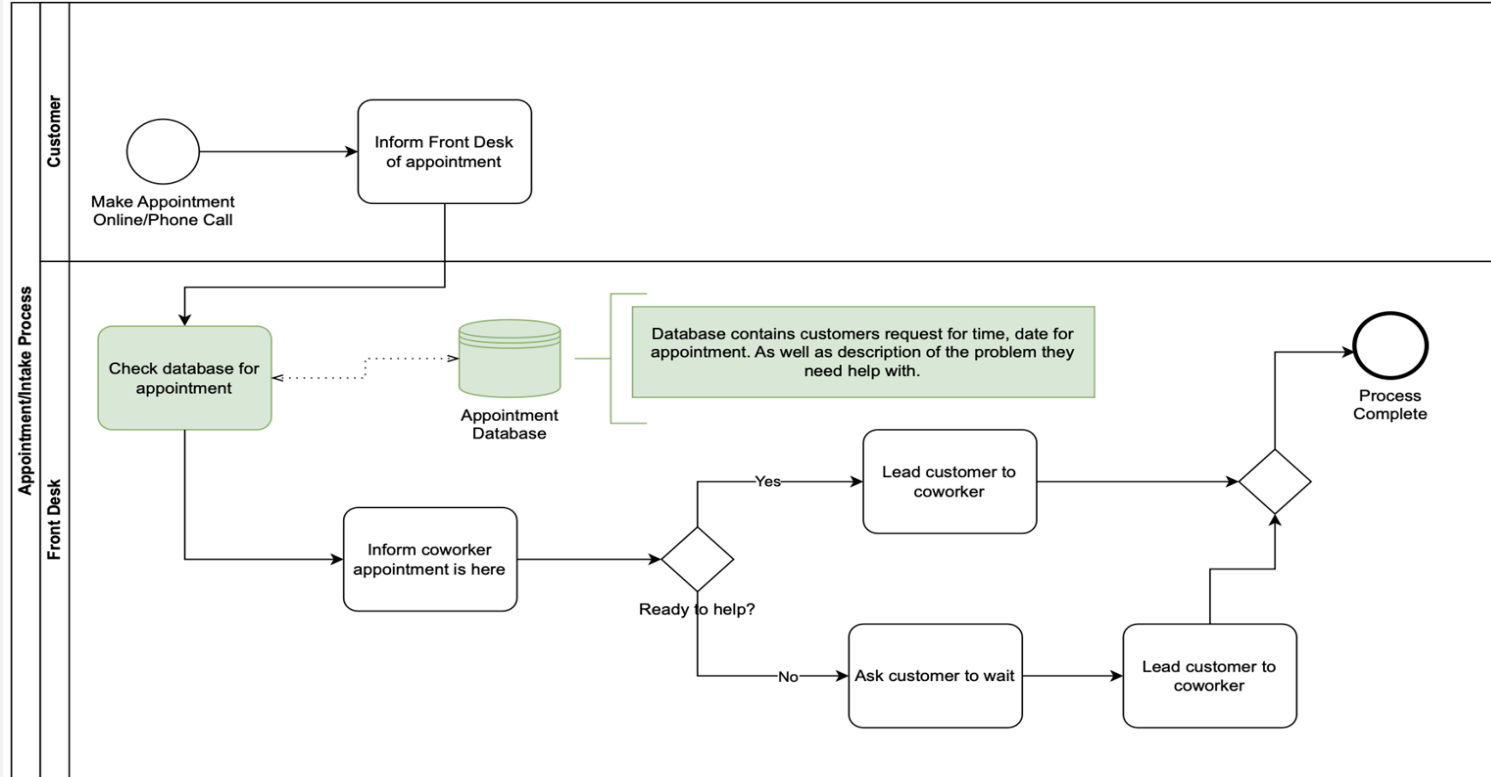
07 Conclusion



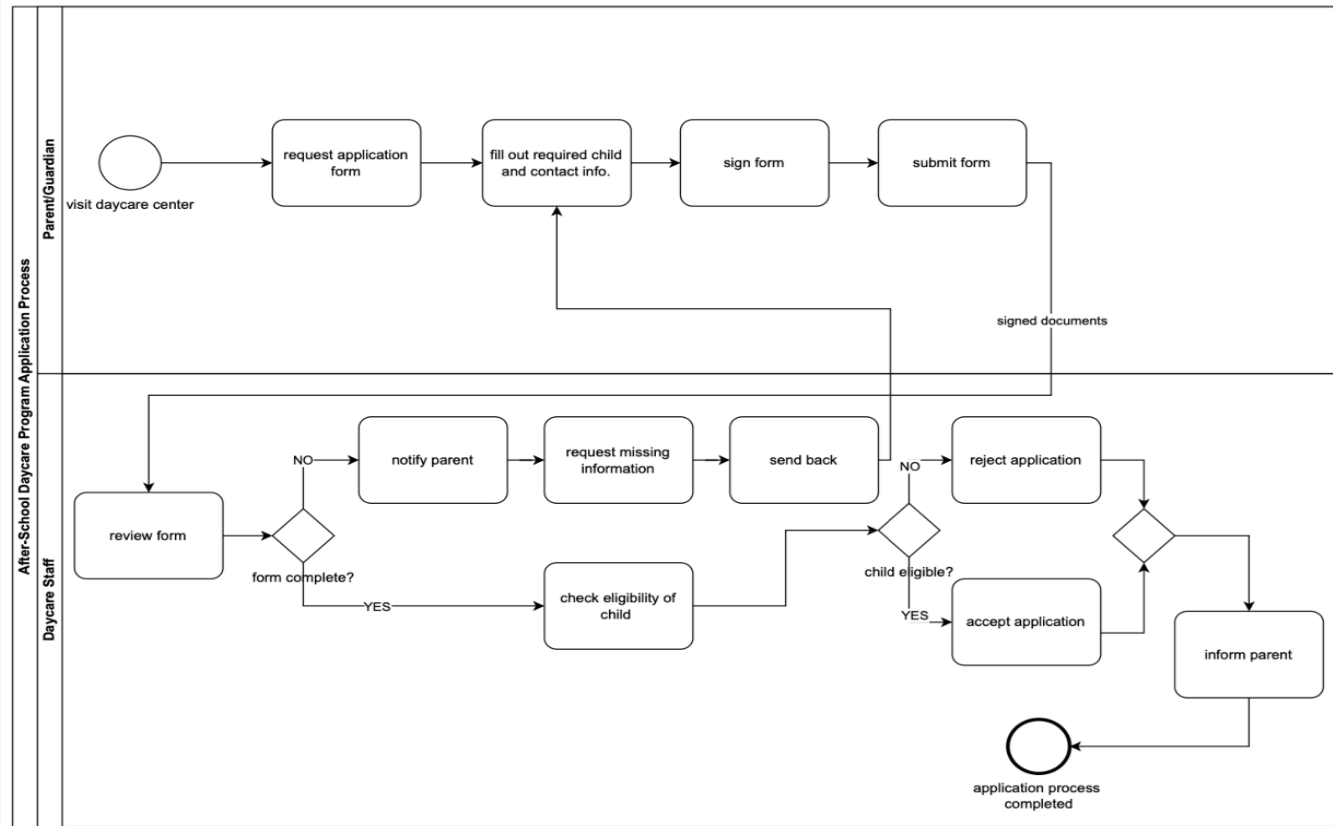
Intake/Appointment Original Process



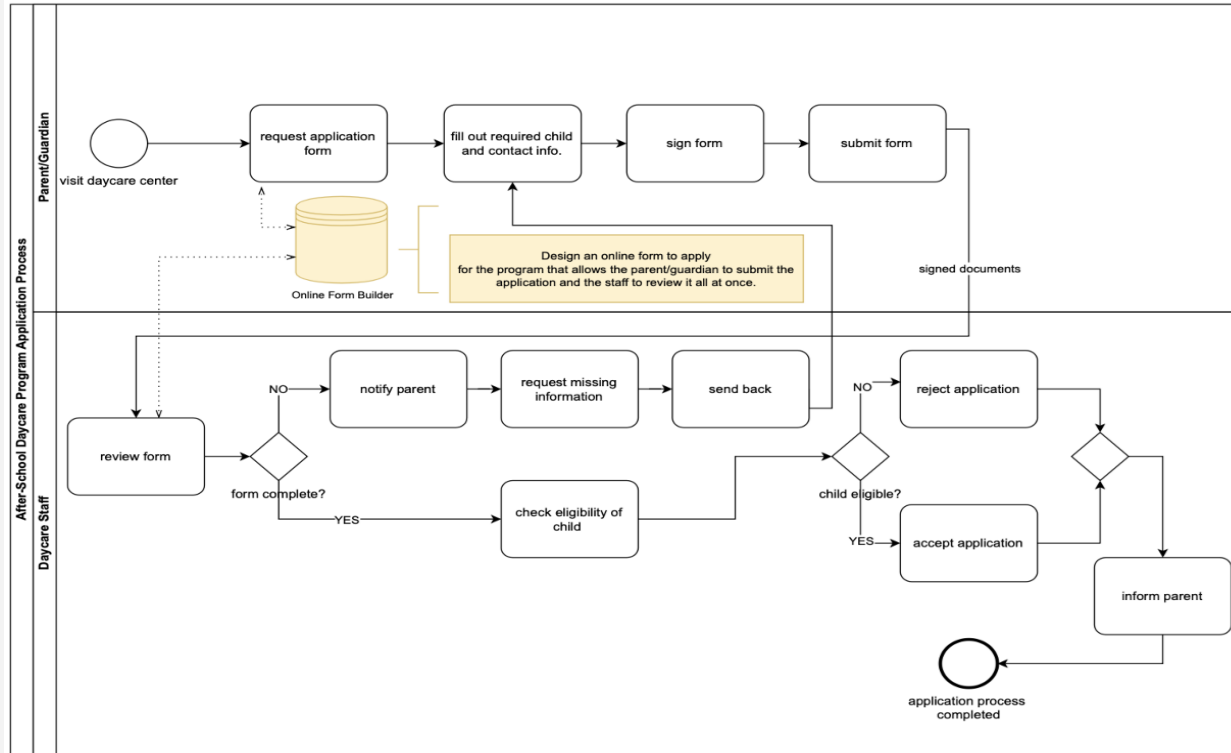
Intake/Appointment Redesign Process



After-School Daycare Program Application Original Process



After-School Daycare Program Application Redesigned Process



Tables

Staff

staff_id	staff_fname	staff_lname	staff_email	staff_phone	staff_languag...	staff_language...	staff_language...
1	Kelvin	Chan	K.Chan@apaccsf.org	5555406713	English		
2	Kirstie	Dutton	K.Dutton@apaccsf.org	5554728930	English	Mandarin	
3	Carrie	Lui	C.Lui@apaccsf.org	5556194852	English	Mandarin	Cantonese
4	Anna	Liao	A.Liao@apaccsf.org	5558042916	English	Mandarin	Cantonese
5	Elizabeth	Lemos	E.Lemos@apaccsf.org	5552854679	English	Spanish	
6	Minky	Fang	M.Fang@apaccsf.org	5557314982	English	Mandarin	
7	Rex	Tabora	R.Taora@apaccsf.org	5556382197	English		

Appointments

appointment...	client_id	staff_id	date	time	reasoning	confirmed	showed_up
1	101	5	2024-10-01	10:40:00	1	1	1
2	54	4	2024-10-01	11:00:00	2	1	1
3	95	4	2024-10-01	10:50:00	3	1	1
4	89	2	2024-10-06	10:00:00	1	1	1
5	23	6	2024-10-06	10:30:00	1	1	1
6	74	2	2024-10-07	12:15:00	2	1	1
7	51	3	2024-10-09	12:15:00	1	1	1
8	38	3	2024-10-09	10:15:00	1	1	1
9	24	3	2024-10-09	12:50:00	1	1	1
10	45	3	2024-10-09	10:45:00	3	1	1
11	53	3	2024-10-12	12:45:00	3	1	1
12	47	4	2024-10-12	10:30:00	1	1	1
13	11	6	2024-10-13	09:00:00	3	1	1
14	116	2	2024-10-14	09:00:00	1	1	1
15	41	4	2024-10-14	12:55:00	2	1	1
16	90	4	2024-10-14	11:00:00	1	0	1
17	76	2	2024-10-14	10:50:00	2	1	1
18	22	3	2024-10-15	10:20:00	4	1	1
19	86	2	2024-10-16	12:00:00	3	1	1
20	26	6	2024-10-17	12:25:00	1	1	1
21	61	5	2024-10-17	10:45:00	3	1	1
22	10	2	2024-10-17	11:30:00	3	1	1
23	56	4	2024-10-17	09:45:00	2	1	1
24	53	3	2024-10-18	09:50:00	2	1	1
25	67	5	2024-10-18	12:35:00	2	1	1
26	59	5	2024-10-19	09:00:00	1	1	1
27	62	5	2024-10-19	09:30:00	1	1	1
28	55	4	2024-10-20	11:15:00	3	1	1
29	27	2	2024-10-21	11:30:00	1	1	1
30	106	5	2024-10-21	09:30:00	2	0	1
31	93	6	2024-10-25	09:45:00	1	1	1
32	20	6	2024-10-25	10:30:00	1	1	1
33	12	4	2024-10-26	12:30:00	2	0	1
34	45	4	2024-10-27	12:25:00	2	1	1
35	63	5	2024-10-27	09:30:00	3	0	0
36	87	5	2024-10-28	11:15:00	1	1	1

Tables cont.

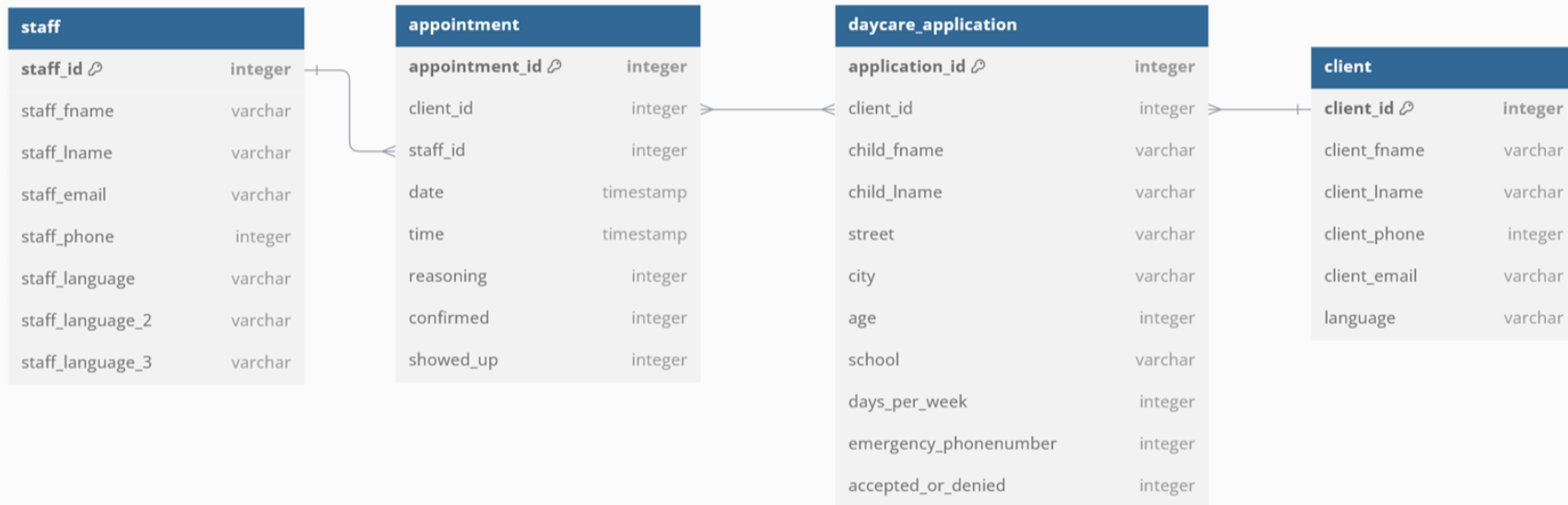
Application

application_...	client_id	child_fname	child_lname	street	city	age	school	days_per_week	emergency_phonenum...	accepted_or_deni...
1	5	Aiden	Fang	Cora	San Francisco	8	Bayshore Elementary	3	5553174921	1
2	6	Bella	Zhi	Cora	San Francisco	9	Bayshore Elementary	2	5554862035	1
3	7	Caleb	Ying	Delta	San Francisco	7	Bayshore Elementary	4	5557285410	1
4	11	Daisy	Li	Cora	San Francisco	8	Bayshore Elementary	3	5558391624	0
5	12	Eli	Hui	Cora	San Francisco	7	Bayshore Elementary	2	5555408293	1
6	13	Fiona	Bo	Teddy Ave	San Francisco	12	Visitation Valley Middle School	4	5552918756	1
7	16	Gavin	Li	Teddy Ave	San Francisco	8	Bayshore Elementary	4	5556724923	1
8	18	Harper	Qiang	Arieta Ave	San Francisco	7	Bayshore Elementary	2	5558246710	1
9	19	Isaac	Hua	Desmond	San Francisco	9	Bayshore Elementary	3	5553196048	1
10	20	Jenna	Ping	Talbert	San Francisco	8	Bayshore Elementary	3	5554532807	1
11	21	Kyle	Lin	Rutland	San Francisco	13	Visitation Valley Middle School	4	5556197032	1
12	22	Lily	Jun	Peabody	San Francisco	7	Bayshore Elementary	5	5557915246	0
13	23	Mason	Ping	Teddy	San Francisco	9	Bayshore Elementary	5	5556435892	1
14	27	Nadia	Hong	Meltra Ct	San Francisco	8	Bayshore Elementary	5	5552108769	0
15	28	Owen	Ming	Schwerin	San Francisco	7	Bayshore Elementary	5	5554973158	1
16	29	Piper	Xia	Teddy Ave	San Francisco	8	Bayshore Elementary	2	5553845721	1
17	30	Quinn	Yu	Meltra Ct	San Francisco	7	Bayshore Elementary	3	5559176504	1
18	31	Riley	Qiang	Meltra Ct	San Francisco	12	Visitation Valley Middle School	5	5552087496	0
19	33	Samuel	Tao	Teddy Ave	San Francisco	8	Bayshore Elementary	4	5556354207	1
20	34	Tessa	Li	Schwerin	San Francisco	9	Bayshore Elementary	5	5559045123	0
21	35	Uma	Xia	Cora	San Francisco	8	Bayshore Elementary	5	5558273145	1
22	36	Vance	Bo	Talbert	San Francisco	10	Visitation Valley Middle School	4	5556827941	0
23	37	Wyatt	Li	Teddy Ave	San Francisco	7	Bayshore Elementary	5	5555036189	1
24	38	Zoe	Li	Schwerin	San Francisco	7	Bayshore Elementary	5	5553967302	0
25	39	Amelia	Yu	Meltra Ct	San Francisco	7	Bayshore Elementary	4	5552678415	0
26	40	Blake	Zhi	Meltra Ct	San Francisco	7	Bayshore Elementary	3	5559102734	1
27	41	Chloe	Li	Teddy Ave	San Francisco	8	Bayshore Elementary	2	5554026819	1
28	42	Dylan	Hui	Schwerin	San Francisco	7	Bayshore Elementary	3	5557385291	0
29	44	Eve	Hui	Talbert	San Francisco	7	Bayshore Elementary	5	5558956147	1
30	45	Felix	Ping	Cora	San Francisco	12	Visitation Valley Middle School	5	5555148902	1
31	46	Grace	Fei	Schwerin	San Francisco	11	Visitation Valley Middle School	4	5557396843	0
32	47	Hugo	Hua	Teddy Ave	San Francisco	11	Visitation Valley Middle School	4	5556814590	1
33	48	Ivy	Zhi	Desmond	San Francisco	10	Visitation Valley Middle School	3	5555207426	0
34	49	Jack	Mei	Schwerin	San Francisco	10	Visitation Valley Middle School	4	5553046792	1
35	50	Kira	Tao	Meltra Ct	San Francisco	11	Visitation Valley Middle School	5	5559215784	1
36	51	Levi	Fei	Teddy Ave	San Francisco	10	Visitation Valley Middle School	5	5557083169	1
37	52	Maya	Ming	Schwerin	San Francisco	11	Visitation Valley Middle School	5	5558527014	0
38	53	Noah	Li	Meltra Ct	San Francisco	12	Visitation Valley Middle School	4	5556723058	1

Client Information

client_id	client_fname	client_lname	client_pho...	client_email	language
1	Zhang	Xian	5554862035	-	Mandarin
2	Wang	Mei	5557285410	wmei@gmail.com	Mandarin
3	Liu	Jun	5558391624	ljun@gmail.com	Mandarin
4	Chen	Hao	5555408293	chao@gmail.com	Mandarin
5	Li	Fang	5552918756	lfang@gmail.com	Mandarin
6	Yang	Zhi	5556724923	yzhi@gmail.com	Mandarin
7	Huang	Ying	5558246710	hying@gmail.com	Mandarin
8	Liu	Tian	5553196048	ltian@gmail.com	Mandarin
9	Zhang	Lian	5554532807	zlian@gmail.com	Mandarin
10	Guo	Xia	5556197032	gxia@gmail.com	Mandarin
11	Zhou	Li	5557915246	zli@gmail.com	Mandarin
12	Wu	Hui	5556435892	whui@gmail.com	Mandarin
13	Song	Bo	5552108769	sbo@gmail.com	Mandarin
14	Yu	Yan	5554973158	yyan@gmail.com	Mandarin
15	Cheng	Ming	5553845721	cming@gmail.com	Mandarin
16	Zhao	Li	5559176504	zli@gmail.com	Mandarin
17	Li	Ming	5552087496	-	Mandarin
18	Sun	Qiang	5556354207	sqiang@yahoo...	Mandarin
19	Tang	Hua	5559045123	thua@yahoo.com	Mandarin
20	Xie	Ping	5558273145	xping@yahoo.com	Mandarin
21	Feng	Lin	5556827941	flin@yahoo.com	Mandarin
22	He	Jun	5555036189	hjun@yahoo.com	Mandarin
23	Wei	Ping	5553967302	wping@yahoo.c...	Mandarin
24	Zhao	Xiu	5552678415	zxliu@yahoo.com	Mandarin
25	Chen	Min	5559102734	cmin@yahoo.com	Mandarin
26	Xu	Nan	5554026819	xnan@yahoo.com	Mandarin
27	Deng	Hong	5557385291	dhong@yahoo.c...	Mandarin
28	Li	Ming	5558956147	lming@yahoo.com	Mandarin
29	Huang	Xia	5555148902	hxia@yahoo.com	Cantonese
30	Tian	Yu	5557396843	tyu@yahoo.com	Cantonese
31	Yang	Qiang	5556814590	yqiang@yahoo...	Cantonese
32	Li	Ying	5555207426	lying@yahoo.com	Cantonese
33	Zhou	Tao	5553046792	-	Cantonese
34	Wu	Li	5559215784	wli@yahoo.com	Cantonese
35	Zhang	Xia	5557083169	zxia@yahoo.com	Cantonese
36	Wang	Bo	5558527014	wbo@gmail.com	Cantonese
37	Zhang	Li	5556723058	zli@gmail.com	Cantonese

Database Diagram



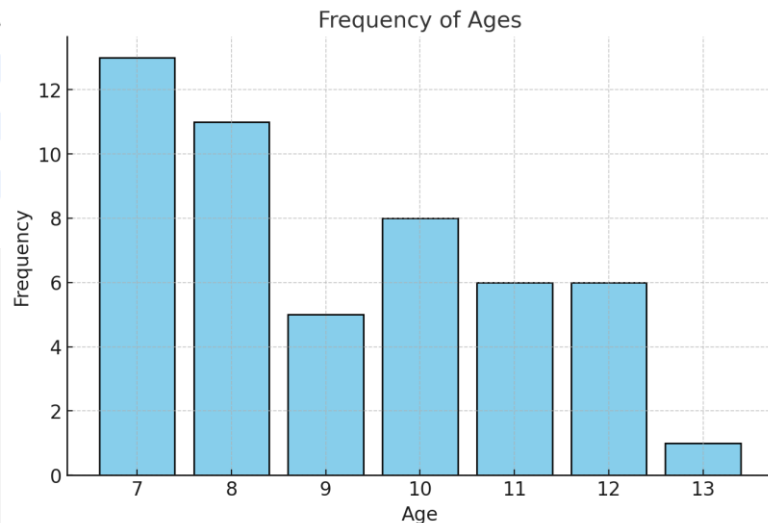
Gaining Insights from Data

- **Mode:** 7 years old
- **Mean:** 9.1 (roughly 9 years old)
 - Total of 50 children at the daycare
 - Focus around the age group of 9 (the average attendee is 9 years old)
 - Special attention to ages 7 and 8
- **Recommendations:**
 - Advertise to parents with children in elementary school
 - Hire staff that would fit the needs of the target age group

```
6 • SELECT AGE , COUNT(AGE) AS Client_Frequency
7 FROM daycare_application
8 GROUP BY AGE
9 ORDER BY AGE ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

AGE	Client_Frequency
7	13
8	11
9	5
10	8
11	6
12	6
13	1



Gaining Insights from Data

- **70% acceptance rate:** most applicants meet the daycare's criteria
- **30% denial rate:** potential areas for improvement in the admissions process
 - Possible reasons: incomplete applications, applicants not meeting specific criteria
- **Recommendations:**
 - Identify key reasons for denial to improve the process
 - Offer resources or guidance to help families meet requirements
 - Revise admission criteria to align with community needs

2. How many applicants are accepted or denied? What is the rate of acceptance and denial?

```
select accepted_or_denied as application_status, count(*) as applicant_count,  
round((count(*) * 100 / sum(count(*)) over()),2) as acceptance_rate  
from daycare_application  
group by accepted_or_denied  
;
```

	application_status	applicant_count	acceptance_rate
	1	35	70.00
	0	15	30.00

Gaining Insights from Data

The highest average age of children attend the daycare 4-5 days a week

- 2 days per week → 8.17 years old
- 3 days per week → 8.60 years old
- 4 days per week → 9.93 years old
- 5 days per week → 9.05 years old

3. What is the average age of the children based on the number of days they plan to attend per week?

```

;
* SELECT Days_Per_Week, AVG(AGE) AS Avg_Age
FROM daycare_application
GROUP BY Days_Per_Week
ORDER BY Days_Per_Week ASC
;

```

Recommendation:

- Prioritize purchasing toys, educational materials, and hire/train specially trained staff to care for children aged 9-10.
- Focus marketing tactics targeted specifically on children aged 9-10

[illegible]

Gaining Insights from Data

- **APACC has frequent visitors for their daycare**
 - Majority of clients visit on a medium basis
(Between 3 and 4 days) → 24 clients
 - The high category is closely right behind
(5 or more days) → 20 clients
 - Few clients visit on a low basis
(2 or fewer days) → 6 clients
- **Recommendation: Allocate resources more efficiently**
 - Improve scheduling and time slots availability
 - Appropriate amount of supplies on hand
 - Preparation of staff availability to assist clients

4. How many daycare clients fall into different categories based on how frequently they visit the daycare throughout the week? The low category being 2 or less visits per week, the medium category being between 3 to 4 visits per week, and the high category being 5 or more visits per week.

```
SELECT Visits_per_Week, Count(*) as Quantity
FROM
(
  SELECT days_per_week,
  CASE
  WHEN days_per_week <= 2
  THEN 'Low'
  WHEN days_per_week BETWEEN 3 AND 4
  THEN 'Medium'
  WHEN days_per_week >= 5
  THEN 'High'
  END AS Visits_per_Week
FROM daycare_application
Order BY Visits_per_Week
) alias
GROUP BY Visits_per_Week
ORDER BY Quantity DESC
;
```

	Visits_per_Week	Quantity
▶	Medium	24
	High	20
	Low	6

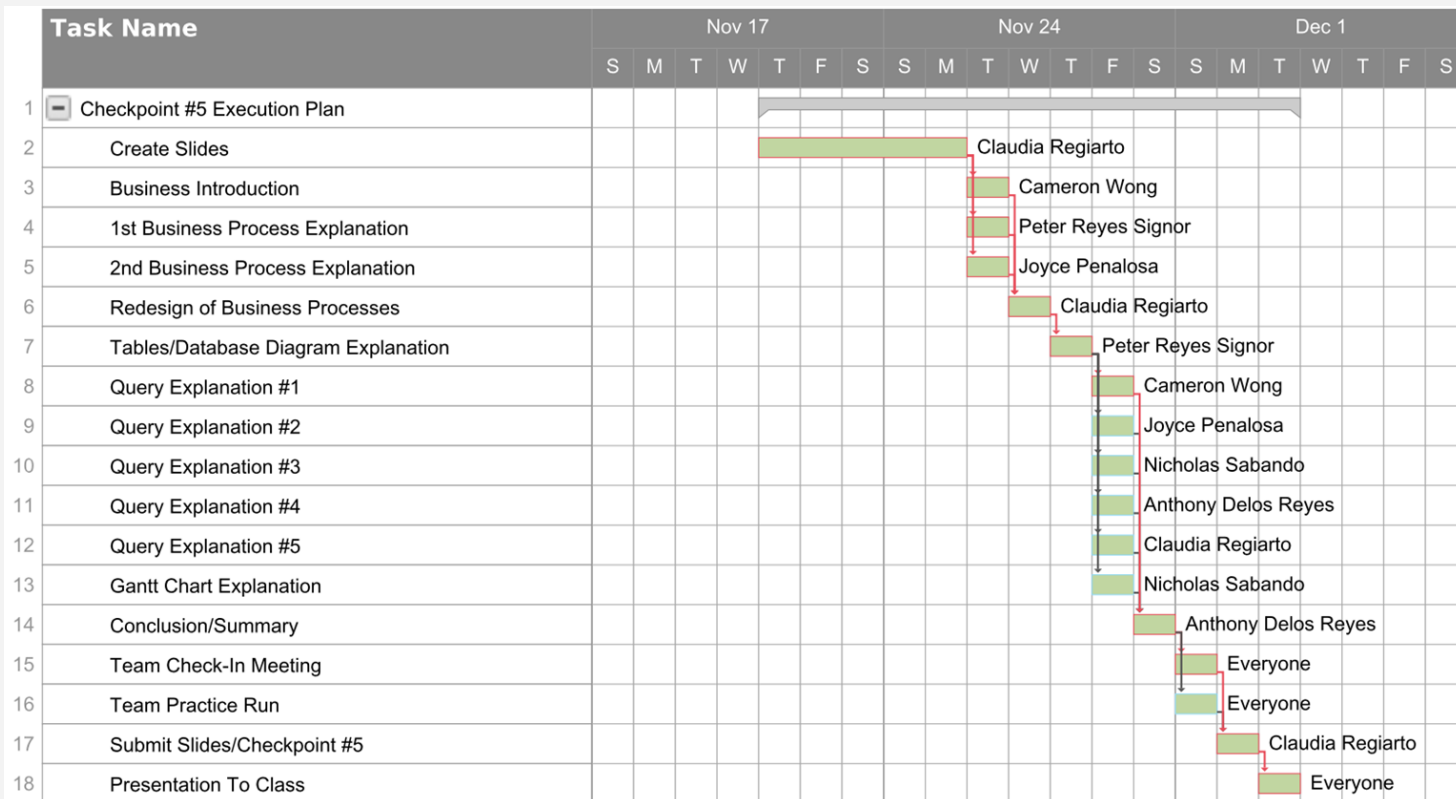
Gaining Insights from Data

- **How many clients fall into each frequency category for specific reasoning**
 - Reasoning #1: English
 - Reasoning #2: Math
 - Reasoning #3: Science
 - Reasoning #4: Social Studies
- **Recommendations:**
 - Optimize marketing strategies
 - Align services to needs

visits_per_day	reasoning	number_of_clie...
High	1	8
High	3	3
High	2	3
High	4	1
Low	2	4
Medium	1	4
Medium	2	4
Medium	3	3
Medium	4	1

```
select visits_per_day, reasoning, count(*) as number_of_clients
from
(
select client_id,
    case
        when days_per_week <= 2 then 'Low'
        when days_per_week between 3 and 4 then 'Medium'
        when days_per_week >= 5 then 'High'
    end as visits_per_day
from daycare_application
) c #alias
join appointment on c.client_id = appointment.client_id
group by visits_per_day, reasoning
order by visits_per_day, number_of_clients desc
;
```

Gantt Chart



Conclusion



1

Goal was to improve client outreach and engagement for families using their services

2

Appointment Redesign → Improve overall efficiency of team's preparation for clients

3

After-School Daycare Redesign → Make form user friendly for clients and easy to review for employees

4

SQL Queries → APACC to have better understanding of their clients

***Thank
You!***

